Loss Landscape Topology Predicts Mitigation Benefit in Continual Learning: A Cross-Dataset Moderation Analysis

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Abstract

We investigate whether topological features of neural network loss landscapes predict resistance to catastrophic forgetting across 19 architectures and 3 datasets (CIFAR-100, CUB-200-2011, NWPU-RESISC45; 57 total configurations). Using persistent homology on 2D cross-sections of loss landscapes (5 independent slices per architecture), we compute H_0 (connected components) and H_1 (loops) persistence via Ripser and GUDHI cubical complexes. Per-dataset leave-one-architecture-out (LOAO) Ridge regression with permutation testing shows topology rescues forgetting prediction on CUB-200 (permutation $p = 0.037$, suggestive) but not RESISC-45 ($p = 0.566$) or CIFAR-100 ($p = 0.295$). The strongest cross-dataset signal is not forgetting prediction but mitigation sensitivity: H_0 predicts EWC benefit on CIFAR-100 ($\rho = 0.76$, $p = 0.0002$) and RESISC-45 ($\rho = 0.86$, $p = 2.4 \times 10^{-6}$) but not CUB-200 ($\rho = 0.31$, $p = 0.19$). A pooled OLS interaction model with clustered bootstrap ($n = 57$, 19 architecture blocks, 5000 iterations) formally confirms that dataset moderates the H_0–EWC benefit relationship (permutation $p = 0.046$). We propose the basin fragmentation hypothesis: H_0 measures landscape fragmentation that determines how much curvature-based regularization prevents inter-basin drift. A WRN-28-k width ladder ($k = 1, 2, 4, 6, 8, 10$) shows H_0 monotonically decreases with width ($\rho = -1.0$) across all 3 datasets, confirmed by cubical persistence ($\rho = 1.0$ agreement with Ripser). Code and results available at `https://github.com/Axion-Deep-Labs/persist-topological-forgetting`.

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1 Introduction

Catastrophic forgetting, the abrupt loss of previously learned knowledge when a neural network is trained on new data, remains a central obstacle in continual learning [McCloskey and Cohen, 1989, French, 1999]. Despite decades of mitigation strategies including elastic weight consolidation [EWC; Kirkpatrick et al., 2017], synaptic intelligence [Zenke et al., 2017], progressive neural networks [Rusu et al., 2016], and replay-based methods [Rebuffi et al., 2017], a fundamental question persists: can we predict, before sequential training begins, how resistant a given architecture will be to catastrophic forgetting, and how much a given mitigation method will help?

Recent work has established that the geometry of loss landscapes encodes meaningful information about generalization [Li et al., 2018, Keskar et al., 2017] and mode connectivity [Garipov et al., 2018]. Separately, topological data analysis (TDA), particularly persistent homology, has emerged as a principled framework for characterizing multi-scale structure in high-dimensional data [Carlsson, 2009]. We bridge these two lines of work by asking: **do topological features of the loss landscape predict forgetting resistance and, critically, mitigation benefit?**

Our contributions are:

1. A cross-architecture, cross-dataset study (19 architectures, 3 datasets, 57 configurations) with a 7-phase experimental pipeline including both Ripser and GUDHI cubical persistence.

2. Evidence that topology's predictive value for forgetting is conditional on dataset: topology rescues prediction on CUB-200 (permutation $p = 0.037$, suggestive) but not on RESISC-45 ($p = 0.566$) or CIFAR-100 ($p = 0.295$, where parameter count dominates).

3. Discovery that H_0 persistence predicts EWC mitigation benefit ($\rho = 0.76$–0.86) on 2 of 3 datasets, formally confirmed by a pooled interaction model with clustered bootstrap (permutation $p = 0.046$).

4. The basin fragmentation hypothesis linking H_0 to regularization sensitivity: fragmented landscapes (high H_0) benefit more from curvature-based regularization that prevents inter-basin drift.

5. A WRN-28-k width ladder (6 widths, $k = 1, 2, 4, 6, 8, 10$) isolating topology from model scale, showing universal H_0 monotonicity ($\rho = -1.0$ vs. width on all 3 datasets).

6. Analysis transparency: the EWC benefit finding emerged from exploratory analysis and is reported as such, with explicit falsification targets.

7. Open-source release of all code, configurations, and results.[1]

2 Related Work

Continual Learning and Catastrophic Forgetting. Strategies for mitigating forgetting broadly fall into regularization-based methods [Kirkpatrick et al., 2017, Zenke et al., 2017], replay-based methods [Rebuffi et al., 2017, Shin et al., 2017], and architecture-based methods [Rusu et al., 2016, Mallya and Lazebnik, 2018]. EWC [Kirkpatrick et al., 2017] penalizes changes to parameters deemed important by the Fisher information matrix, effectively anchoring the network near the Task A minimum. Our work is orthogonal to method development: rather than proposing a new mitigation strategy, we investigate whether pre-training landscape properties predict how much a given mitigation method will help.

Loss Landscape Analysis. Li et al. [2018] introduced filter-normalized random directions for meaningful cross-architecture visualization of loss surfaces. Keskar et al. [2017] linked sharpness (largest Hessian eigenvalue) to generalization. Garipov et al. [2018] demonstrated low-loss paths between independently trained minima, establishing mode connectivity as a key landscape property. We adopt filter-normalized 2D cross-sections following Li et al. [2018] and compute topological invariants on the resulting surfaces.

[1] `https://github.com/Axion-Deep-Labs/persist-topological-forgetting`

TDA in Machine Learning. Persistent homology has been applied to neural network analysis in several contexts: characterizing decision boundaries [Ramamurthy et al., 2019], analyzing weight-space topology during training [Rieck et al., 2019], studying activation patterns [Naitzat et al., 2020], and providing stable vectorizations via persistence images [Adams et al., 2017]. Our work differs in applying persistent homology directly to loss landscape *surfaces* rather than to weight trajectories or activation manifolds. Specifically, we use both Vietoris-Rips (via Ripser) and cubical complex (via GUDHI) persistence on 2D loss grids to extract H_0 and H_1 features, and we relate these to continual learning dynamics rather than single-task generalization.

3 Methodology

3.1 Overview

Our experimental protocol consists of seven phases, each producing data consumed by subsequent phases:

P1. Train Task A to convergence.

P2. Compute 5 Ripser landscape slices per architecture.

P2c. Compute cubical persistence via GUDHI.

P3. Measure forgetting under naive, EWC, and cosine LR schedules.

P4. Per-dataset correlation and diagnostic analysis.

P5. Per-dataset LOAO predictive model with permutation test.

P6. Pooled OLS interaction model with clustered bootstrap.

3.2 Datasets

We evaluate on three datasets selected for diversity in domain and granularity (Table 1). All images are resized to 32×32 pixels for cross-architecture consistency.

CIFAR-100 provides a standard benchmark with coarse-grained categories. CUB-200-2011 [Wah et al., 2011] introduces fine-grained visual recognition where inter-class differences are subtle. NWPU-RESISC45 [Cheng et al., 2017] tests generalization to a non-natural-image domain (remote sensing). Standard augmentation (random crop with padding, horizontal flip) is applied during training; no augmentation at test time.

Table 1: Datasets used in the study. Each dataset is split into two disjoint class subsets (Task A / Task B) for sequential training.

Dataset	Domain	Classes	Split
CIFAR-100	Natural images	100	50 / 50
CUB-200-2011	Fine-grained birds	200	100 / 100
NWPU-RESISC45	Satellite scenes	45	23 / 22

Table 2: Architectures evaluated. The WRN-28-k width ladder isolates capacity effects within a single architecture family.

Architecture	Params	Type
ViT-Tiny	0.3M	Transformer
WRN-28-1	0.4M	WRN ladder
MobileNet-V3-S	1.1M	CNN
ShuffleNet-V2	1.3M	CNN
WRN-28-2	1.5M	WRN ladder
ViT-Small	2.2M	Transformer
MLP-Mixer	2.3M	MLP
RegNet-Y-400MF	4.0M	CNN
EfficientNet-B0	4.1M	CNN
WRN-28-4	5.9M	WRN ladder
DenseNet-121	7.1M	CNN
ResNet-18	11.2M	CNN
WRN-28-6	13.2M	WRN ladder
VGG-16-BN	14.8M	CNN
WRN-28-8	23.4M	WRN ladder
ResNet-50	23.7M	CNN
ConvNeXt-Tiny	27.9M	Modern CNN
WRN-28-10	36.5M	WRN ladder
ResNet-18 Wide	44.7M	CNN

3.3 Architectures

We evaluate 19 architectures spanning five computational paradigms (Table 2): 14 diverse architectures covering CNNs, Vision Transformers, and MLP-based designs, plus a WRN-28-k width ladder with $k \in \{1, 2, 4, 6, 8, 10\}$. The width ladder isolates the effect of model capacity on topology while holding architecture family constant. Parameter counts range from 0.3M (ViT-Tiny) to 44.7M (ResNet-18 Wide).

All models are trained from scratch with a fixed global seed. CNN-based architectures are adapted for 32×32 input with 3×3 initial convolutions (stride 1); ViT variants use patch size 4.

3.4 Loss Landscape Topology

We compute persistent homology on 2D cross-sections of the loss landscape around each converged minimum $\boldsymbol{\theta}^*$, using two complementary methods.

Landscape sampling. For each architecture and each of 5 independent random seeds, we generate two random directions $\mathbf{d}_1, \mathbf{d}_2$ in parameter space with *filter normalization* [Li et al., 2018]: for each convolutional filter (or weight matrix) i, rescale $\mathbf{d}^{(i)}$ so that $\|\mathbf{d}^{(i)}\| = \|\boldsymbol{\theta}^{*(i)}\|$. The loss is evaluated on a uniform 50×50 grid over $[-1, 1]^2$:

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}(\boldsymbol{\theta}^* + \alpha\,\mathbf{d}_1 + \beta\,\mathbf{d}_2)\,. \tag{1}$$

Ripser persistence. Let $V = \{v_{ij}\}$ denote the $50 \times 50 = 2{,}500$ grid vertices. Assign each vertex a filtration value $f(v_{ij}) = \mathcal{L}(\alpha_i, \beta_j)$. Connect vertices by 8-adjacency (cardinal and diagonal neighbors). Each edge receives the lower-star weight:

$$w(u, v) = \max\bigl(f(u),\ f(v)\bigr). \tag{2}$$

We pass the weighted graph to Ripser [Bauer, 2021] and compute H_0 (connected components) and H_1 (independent loops) persistence diagrams.

GUDHI cubical persistence. As a validation baseline, we compute sublevel-set persistence directly on the 50×50 loss grid using GUDHI's `CubicalComplex`. This avoids the graph-construction step and operates on the raw scalar field.

Cross-method agreement. H_1 total persistence from Ripser and GUDHI shows Spearman $\rho = 1.0$ agreement on all 3 datasets, confirming that the topological measurements are method-independent.

For each homology dimension k, we record total persistence $\mathrm{Pers}_k = \sum_i (d_i - b_i)$, feature count, and maximum lifetime. Using 5 independent slices per architecture (seeds logged for reproducibility) substantially reduces sampling variance compared to single-slice designs.

3.5 Forgetting Measurement

Starting from the Task A checkpoint, we expand the final classification layer and train on Task B under three conditions:

1. **Naive:** SGD with fixed learning rate 0.01.
2. **EWC:** Naive training plus elastic weight consolidation [Kirkpatrick et al., 2017] with diagonal Fisher information matrix ($\lambda = 1000$).
3. **Cosine LR:** Naive training with cosine annealing learning rate schedule.

Task A test accuracy is evaluated at steps $\{10, 25, 50, 100, 250, 500, 1000, 5000\}$. Our primary retention metric is:

$$\text{ret@}k = \frac{\text{acc}_A(\text{step} = k)}{\text{acc}_A(\text{step} = 0)}. \tag{3}$$

We also compute the area under the retention curve (AURC) via trapezoidal integration over the early evaluation steps (steps 0–500), denoted early_aurc. The EWC benefit is defined as the absolute improvement in retention:

$$\text{EWC\_benefit} = \text{ret}_{\text{EWC}} - \text{ret}_{\text{naive}}. \tag{4}$$

3.6 Statistical Analysis

Per-dataset analysis (Phases 4–5). For each dataset independently, we compute:

- Spearman rank correlations between topology/params and forgetting metrics, with Bonferroni correction across 12 tests (adjusted $\alpha = 0.004$).
- Partial correlations controlling for parameter count.
- Within-WRN-ladder analysis (6 widths, fixed architecture family).
- LOAO Ridge regression: for each held-out architecture, fit a Ridge model on the remaining $n - 1$ architectures and predict the held-out retention. Nested alpha selection via inner LOO. We compare three feature sets: params-only (log params), topology-only (H_0, H_1, feature counts), and params + topology.
- Permutation test: 1000 shuffles of topology columns, testing whether topology improves prediction beyond params.
- Matched-dimensionality null control: 1000 random feature draws matching topology dimensionality, to verify that improvement is not an artifact of added degrees of freedom.

Pooled interaction analysis (Phase 6). To formally test whether dataset moderates the topology–forgetting relationship, we pool all 57 observations (19 architectures × 3 datasets) and fit OLS models with interaction terms, using CIFAR-100 as the reference dataset:

$$\begin{aligned} \text{M0:}\quad Y &= \beta_0 + \beta_1 \hat{p} + \beta_2 D_C + \beta_3 D_R \\ &\quad + \beta_4 \hat{p} D_C + \beta_5 \hat{p} D_R + \epsilon \end{aligned} \tag{5}$$

$$\text{M1:}\quad Y = \text{M0} + \gamma_1 \tilde{H}_0 + \gamma_2 \tilde{H}_0 D_C + \gamma_3 \tilde{H}_0 D_R + \epsilon \tag{6}$$

where $\hat{p} = \log(\text{params})$ globally standardized, $\tilde{H}_0$ is H_0 z-scored within each dataset, D_C and D_R are indicator variables for CUB-200 and RESISC-45 respectively. H_0 is z-scored within dataset to remove cross-dataset scale differences while preserving within-dataset rank information.

Inference. We use clustered bootstrap (5000 iterations, resampling 19 architecture blocks of 3) to obtain 95% confidence intervals that account for the non-independence of observations from the same architecture across datasets. Permutation testing (1000 iterations, H_0 shuffled within dataset) provides a distribution-free significance test. The primary hypothesis test is a block test of $\{\gamma_1, \gamma_2, \gamma_3\}$ (does adding H_0 and its interactions improve M0?). The secondary test targets $\{\gamma_2, \gamma_3\}$ conditional on γ_1 (does the H_0 effect vary across datasets?). Variance inflation factors (VIF) are computed for all predictors; all fall below 3.5 in the z-scored specification.

Robustness checks include: (i) a reduced model omitting $\hat{p} \times D$ interactions, (ii) raw H_0 (not z-scored) sensitivity analysis, and (iii) alternative outcome variables.

4 Results

4.1 Per-Dataset Forgetting Prediction

Table 3 presents the LOAO Ridge regression results for each dataset. Topology's predictive value varies dramatically across datasets.

Table 3: Per-dataset LOAO Ridge regression results. Topology rescues prediction on CUB-200 but not the other two datasets. Bold indicates suggestive or significant results.

Dataset	Outcome	Params ρ	+Topo ρ	Perm. p
CIFAR-100	ret@100	0.43	0.30	0.295
CUB-200	ret@10	−0.92	**0.34**	**0.037**
RESISC-45	ret@100	−0.32	−0.33	0.566

CIFAR-100. Parameter count dominates forgetting prediction ($\rho = -0.76$, $p = 0.0002$, survives Bonferroni). Adding topology does not improve the params-only model (permutation $p = 0.295$). Topology is redundant when scale already explains the outcome.

CUB-200. Parameters alone fail ($\rho = -0.27$, $p = 0.27$) and, used as the sole LOAO predictor, produce

Table 4: Spearman correlation between H_0 persistence and EWC benefit ($\text{ret}_{\text{EWC}} - \text{ret}_{\text{naive}}$). Bold: survives Bonferroni ($\alpha = 0.004$).

Dataset	$\rho(H_0, \text{EWC benefit})$	p-value
CIFAR-100	**0.76**	**0.0002**
RESISC-45	**0.86**	2.4×10^{-6}
CUB-200	0.31	0.19

a correlation of $\rho = -0.92$ with observed retention (wrong direction: the model systematically predicts high retention for architectures that forget and vice versa). Adding topology rescues prediction: LOAO $\rho = 0.34$, permutation $p = 0.037$. Topology-only achieves $\rho = 0.33$ with MAE = 0.147, outperforming params-only. The matched-dimensionality control exceeds the 95th percentile, and MAE reduction is 17.5%. However, $p = 0.037$ does *not* survive Bonferroni correction across 3 datasets (adjusted $\alpha = 0.0167$). We report this as suggestive.

RESISC-45. Neither parameters ($\rho = -0.29$, $p = 0.22$) nor topology (permutation $p = 0.566$) predict forgetting. The signal is absent on this dataset.

4.2 EWC Benefit: The Cross-Dataset Signal

While topology's relationship to raw forgetting is dataset-dependent, we find a more consistent signal when the outcome is *mitigation benefit* rather than forgetting itself. Table 4 shows H_0 persistence correlates with EWC benefit on 2 of 3 datasets.

This is the most stable cross-dataset signal in our study. Architectures with more fragmented loss landscapes (higher H_0) benefit more from EWC regularization. The effect is strong on CIFAR-100 and RESISC-45 but absent on CUB-200, suggesting that the mechanism linking landscape topology to mitigation benefit depends on task characteristics.

4.3 Pooled Interaction Analysis

To formally test whether dataset moderates the H_0 effect, we fit the pooled OLS interaction model (Equations 5–6) on all 57 observations. Table 5 presents the main results.

The primary result is the EWC benefit (AURC) model: adding H_0 and its dataset interactions improves R^2 from 0.502 to 0.587, with permutation $p = 0.046$ for the block test. The ret@100 model reaches $p = 0.035$ as a robustness check, while the primary forgetting outcome (ret@10) does not reach significance ($p = 0.196$).

Table 5: Pooled interaction model results. ΔR^2 is the improvement from adding H_0 terms (M1 vs. M0). Block p tests $\{\gamma_1, \gamma_2, \gamma_3\}$ jointly. Bold: $p < 0.05$.

Outcome	R^2_{M0}	R^2_{M1}	ΔR^2	Block p	Verdict
EWC ben. (AURC)	0.502	0.587	**0.085**	**0.046**	Significant
ret@10	0.179	0.254	0.075	0.196	NS
ret@100	0.297	0.423	**0.127**	**0.035**	Significant

Table 6: Per-dataset partial effect of $\tilde{H}_0$ (z-scored within dataset) with 95% clustered bootstrap CIs. Bold: CI excludes zero.

Outcome	Dataset	Effect	95% CI
ret@10	CIFAR-100	−0.001	[−0.486, +0.073]
	CUB-200	**−0.123**	**[−0.183, −0.046]**
	RESISC-45	−0.021	[−0.264, +0.083]
EWC ben. (AURC)	CIFAR-100	**+0.016**	**[+0.005, +0.062]**
	CUB-200	+0.002	[−0.008, +0.013]
	RESISC-45	**+0.007**	**[+0.004, +0.012]**

Per-dataset partial effects. Table 6 shows the per-dataset H_0 partial effects with 95% clustered bootstrap confidence intervals.

For ret@10, only the CUB-200 partial effect excludes zero from its CI, consistent with the per-dataset LOAO result. For EWC benefit, CIFAR-100 and RESISC-45 both exclude zero, while CUB-200 does not, consistent with the per-dataset Spearman correlations in Table 4. Figure 1 visualizes the moderation pattern.

All VIF values fall below 3.5 in the z-scored specification. Robustness checks confirm consistent inference: the reduced model (omitting $\hat{p} \times D$ interactions) yields $p = 0.031$ for both ret@100 and EWC benefit. The raw H_0 (not z-scored) sensitivity analysis produces the same qualitative conclusions.

4.4 WRN Width Ladder

The WRN-28-k width ladder ($k = 1, 2, 4, 6, 8, 10$) provides a controlled test of how model capacity affects topology within a single architecture family. Table 7 summarizes the key finding.

H_0 persistence monotonically decreases with network width ($\rho = -1.0$ vs. parameter count) on CIFAR-100, CUB-200, and RESISC-45. This is consistent with the intuition that wider networks have smoother loss landscapes with fewer discon-

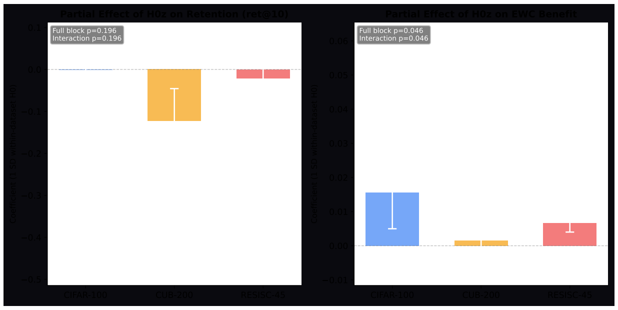

Figure 1: Pooled interaction model results. Left: H_0 vs. EWC benefit by dataset. Right: per-dataset partial effects with 95% clustered bootstrap CIs. The H_0–EWC benefit relationship is strong on CIFAR-100 and RESISC-45 but absent on CUB-200, demonstrating dataset moderation (permutation $p = 0.046$).

Table 7: WRN-28-k width ladder: H_0 monotonically decreases with width on all 3 datasets. Cubical (GUDHI) and Ripser H_1 show perfect agreement.

Property	ρ vs. params	Datasets
H_0 vs. width	-1.0	All 3
Cubical vs. Ripser H_1	$+1.0$	All 3

nected basins. Cubical and Ripser H_1 values show perfect rank agreement ($\rho = 1.0$), confirming that the topological measurement is reliable and method-independent.

The width ladder demonstrates that the topological measurement itself is universal and monotonic with capacity. It is the *relationship* between topology and forgetting/mitigation that varies by task, not the topology-capacity link.

5 Basin Fragmentation Hypothesis

The results in Sections 4.2–4.4 motivate a mechanistic hypothesis linking loss landscape topology to mitigation benefit.

H_0 persistence in the sublevel-set filtration counts connected components and their relative depths. High H_0 corresponds to a *fragmented* landscape with many disconnected low-loss basins separated by high-loss barriers. We propose:

> **Basin Fragmentation Hypothesis.** H_0 measures landscape fragmentation that determines how much curvature-based regularization (e.g., EWC) helps by preventing inter-basin drift during sequential training.

The mechanism is as follows. When the loss landscape is fragmented (high H_0), naive sequential training on Task B can push parameters across basin boundaries, causing them to settle in a different basin that encodes Task B at the expense of Task A. EWC penalizes deviations from the Task A minimum in proportion to Fisher information, effectively constraining the optimization trajectory to remain near the original basin. This constraint is maximally beneficial when there are many basins to drift between (high H_0) and minimally beneficial when the landscape is already smooth with a single broad basin (low H_0), because EWC then addresses a nonexistent problem.

Three lines of evidence support this hypothesis:

1. H_0 predicts EWC benefit on CIFAR-100 ($\rho = 0.76$) and RESISC-45 ($\rho = 0.86$), with the pooled interaction model confirming moderation at $p = 0.046$.

2. The WRN width ladder shows H_0 decreases monotonically with width ($\rho = -1.0$), consistent with wider networks having smoother, less fragmented landscapes.

3. CUB-200 shows no H_0–EWC benefit relationship ($\rho = 0.31$, $p = 0.19$), suggesting that fine-grained recognition may involve *feature-level* interference rather than basin drift, a mechanism that EWC-style regularization addresses differently.

This hypothesis is tentative. A causal test would require a landscape-aware regularization intervention: artificially increasing landscape fragmentation and testing whether EWC benefit increases correspondingly. We leave this to future work.

6 Analysis Path Transparency

The original hypothesis for this study targeted topology as a direct forgetting predictor, motivated by preliminary CIFAR-100 results suggesting H_1 correlation with retention. The expanded study proceeded in temporal order: CIFAR-100 ran first, revealing that parameter count dominates forgetting prediction and H_1 drops to non-significance after partialing out params. CUB-200 ran second, showing topology rescues prediction where params fail (permutation $p = 0.037$). RESISC-45 ran third, producing a null result that falsified the simple "hard tasks need topology" framing. The EWC benefit finding emerged from Phase 4 diagnostic analysis, specifically the correlation of H_0 with EWC retention minus naive retention, and was not part of the original study design.

The pooled interaction model (Phase 6) was designed after observing the per-dataset EWC pattern. Accordingly, the EWC moderation finding ($p = 0.046$) should be interpreted as a *discovery* requiring pre-registered replication, not as confirmatory evidence.

7 Limitations

1. **Small-scale models only.** All 19 architectures are under 45M parameters. Production continual learning systems use models with 100M–7B+ parameters. Whether the topological signal persists at that scale is genuinely unknown and constitutes the primary open research question. Computing persistent homology on large parameter spaces introduces superpolynomial computational barriers that may require novel distributed algorithms and supercomputer resources.

2. **Sample size.** 19 architectures provide moderate statistical power. Per-dataset analyses operate on $n = 19$ (or $n = 6$ for the WRN ladder), limiting detection of weak effects. Robust claims require 50–100+ architectures.

3. **Two-task sequences.** We test only a single Task A $\rightarrow$ Task B transition. Real continual learning involves 10–100+ sequential tasks. Topology–forgetting dynamics over long task sequences are entirely unexplored and may exhibit qualitatively different behavior.

4. **Single mitigation method.** We test only EWC. If synaptic intelligence [Zenke et al., 2017] or PackNet [Mallya and Lazebnik, 2018] show no H_0 correlation, the finding is EWC-specific rather than a general principle about curvature-based regularization.

5. **2D landscape projections.** The 5 independent slices per architecture mitigate but do not eliminate sampling variance from projecting a high-dimensional surface onto 2 dimensions. At production scale, slice fidelity is an open mathematical question.

6. **Small-image datasets.** All images are 32×32. Whether the signal holds on large-scale benchmarks (ImageNet), natural language tasks, or domain-specific data (medical imaging) is unknown.

7. **Borderline p-values.** The EWC moderation result ($p = 0.046$) is borderline. The CUB-200 forgetting result ($p = 0.037$) does not survive Bonferroni correction across 3 datasets. The ret@100 pooled result ($p = 0.035$) was not the pre-specified primary outcome.

8. **Exploratory EWC finding.** As discussed in Section 6, the EWC benefit analysis was not pre-registered.

Falsification targets. We identify four results that, if obtained in follow-up work, would undermine the basin fragmentation hypothesis:

1. Synaptic intelligence benefit shows no H_0 correlation, implying the finding is EWC-specific rather than a general regularization phenomenon.

2. Adding 10+ architectures eliminates the CUB-200 topology–forgetting signal, implying the $p = 0.037$ result was a small-sample artifact.

3. Sharpness-aware minimization (SAM) changes H_0 without changing EWC benefit, breaking the proposed causal link between fragmentation and mitigation.

4. Cubical persistence disagrees with Ripser on the moderation result, implying method dependence in the primary finding.

8 Conclusion

Across 19 architectures and 3 datasets (57 configurations), we find that loss landscape topology is a conditional, not universal, predictor of continual learning dynamics. The primary contribution of this work is a formal moderation test demonstrating that dataset moderates the relationship between H_0 persistence and EWC mitigation benefit (pooled interaction permutation $p = 0.046$). H_0 predicts EWC benefit on CIFAR-100 ($\rho = 0.76$) and RESISC-45 ($\rho = 0.86$) but not CUB-200 ($\rho = 0.31$). The basin fragmentation hypothesis provides a plausible mechanism: fragmented landscapes (high H_0) benefit more from curvature-based regularization that prevents inter-basin drift. The WRN-28-k width ladder confirms that H_0 monotonically decreases with capacity ($\rho = -1.0$ on all 3 datasets), establishing the topology-capacity link as universal even when the topology-forgetting link is not.

Critically, these results constitute a *preliminary proof-of-concept* on small-to-medium models (0.3M–44.7M parameters) and small-image datasets. Whether the topological signal survives at production scale remains genuinely unknown and constitutes the primary open research question. Scaling persistent homology to models with hundreds of millions or billions of parameters introduces fundamental computational barriers: Ripser complexity is superpolynomial in the number of sampled points, and it is unclear whether the 2D subsampling strategy retains fidelity in high-dimensional parameter spaces. Systematic investigation of these questions requires HPC resources (e.g., NSF ACCESS supercomputer allocation) and should: (i) test whether the signal persists on production-scale models (100M–7B+ parameters), which may require novel distributed PH algorithms; (ii) extend to long task sequences (10–100+ sequential tasks) where topology evolution may become chaotic; (iii) evaluate multiple continual learning methods (SI, PackNet, replay, adapter-based) to determine whether the finding generalizes beyond EWC; (iv) scale to 50–100+ architectures for statistical power sufficient to survive strict multiple-comparison correction; and (v) pre-register the moderation test for confirmatory replication.

All code, configurations, and results are available at `https://github.com/Axion-Deep-Labs/persist-topological-forgetting`.

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